جامعة التقنية والعلوم التطبيقية
University of Technology and Applied Sciences
إبراء
Ibra

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University Queue Management Website

A PROJECT REPORT

Submitted in partial fulfillment of the requirement for the

Diploma 2nd Year – [Specialization]

By

UNDER THE GUIDANCE OF

Supervisor name:

BONAFIDE CERTIFICATE

جامعة التقنية والعلوم التطبيقية
University of Technology and Applied Sciences
إبراء
Ibra

This is to certify that the course project entitled ***University Queue Management Website*** **is** an approved work done by

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In partial fulfillment for the requirement of the level Bachelor – Information Security, Information Technology Department of the University of Technology and Applied Science – Ibra, for the school year 2022 – 2023.

Course Project Supervisor **Head of Department – IT**

Approved and submitted during the final project presentation held on _____

EXAMINER 1 **EXAMINER 2**

ABSTRACT

The Department of Student Affairs and Registration at the University of Technology and Applied Sciences Ibra is facing several problems. It is expected to serve clients like staff and students, are intended to have queues managed by a queue management system. [1] This website presents the design of an eco-friendly and more efficient web-based queue management system. Through a Graphical User Interface (GUI), the system enables users to obtain tickets on their mobile devices, doing away with the need for paper. Students who live far from the service center can benefit from the system's notification of approaching turns. Enhancing customer (student/staff) satisfaction and service effectiveness is the system's goal. When students and staff need to access different services, many universities struggle with lengthy lines at their service centers. Both customers and employees may become frustrated and stressed out by prolonged wait times. Universities therefore require a strong queue management system that can shorten wait times and enhance customer satisfaction. This website suggests a web-based queue management system that enables users to get tickets using their smartphones, get alerts when their turns are about to happen, and rate and review their overall experience. To optimize their service delivery, service providers can also make educated decisions by using the system to track and evaluate customer behavior and queue status. The creation of an automated queue management system that can improve client satisfaction and service effectiveness is the goal of this design. A queue is a linear data structure which follows a particular order in which the operations are performed. The order may be **FIFO.**

ACKNOWLEDGEMENT

A waiting line occurs within a system when customers arrive for a product or service, wait in line if the service is not immediate, and leave the queue once they have been served. The queuing theory provides a foundation for determining the most optimal solution for managing queues. By definition, a queue refers to a line of people or vehicles waiting in succession for their turn to be attended to or to proceed [3]. In busy environments where many people are queuing, a numbered ticket system is implemented. When someone arrives, they take a number and wait in line accordingly. A Queue Management System is a software solution designed for managing customer queues, collecting feedback, monitoring real-time information, and streamlining service delivery. Currently, there are three implementations of queues:

1. Standard Linear Queues: This form of queuing represents the normal or standard queue system where each service desk or cashier has a separate line [3].
2. Single Line Queues: Also referred to as a Call Forward System, a single line groups customers and then feeds them to multiple cashiers or service areas. Often, Individual service stations are allocated to different service personnel parts of one long counter or desk [3].
3. Dispersed or “Digital” Queues: This type of queue management disperses waiting lines by offering a ticketing management system [3].

The sequencing of customers for service within queues determines the order in which they are served. Various disciplines exist within this category:

1. First Come, First Served (FCFS): This mode is commonly used in real-world scenarios, such as bank tellers, where customers are served in the order of their arrival.

2. Last Come, First Served (LCFS): In this mode, customers are served in reverse order, with the most recent arrival receiving service first.

3. Priority Discipline (PD): Customers are categorized based on priority levels, and each category is given different service priorities.

Traditionally, queues typically follow either FCFS or LCFS, and any attempts to modify the traditional queue structure by cutting through lines often result in social friction [4]. Consequently, queues often experience long waiting times, not only due to inadequate service distribution among staff but also because of missed or improperly called tickets, leading to unnecessary time wastage.

DECLARATION

Diploma 3 students with specialization in IT and with a course project proposal entitled ***University Queue Management Website*.** Have submitted a complete report. All information in the document have been properly cited and reference have been mentioned in our documentation.

PLAGIARISM REPORT (LESS THAN 30%)
(TURNITIN)

CONTENTS

I. Abstract

1. The Department of Student Affairs and Registration at the University of Technology and Applied Sciences Ibra is facing several problems. It is expected to serve clients like staff and students, are intended to have queues managed by a queue management system. [1] This website presents the design of an eco-friendly and more efficient web-based queue management system. Through a Graphical User Interface (GUI), the system enables users to obtain tickets on their mobile devices, doing away with the need for paper. Students who live far from the service center can benefit from the system's notification of approaching turns. Enhancing customer (student/staff) satisfaction and service effectiveness is the system's goal. When students and staff need to access different services, many universities struggle with lengthy lines at their service centers. Both customers and employees may become frustrated and stressed out by prolonged wait times. Universities therefore require a strong queue management system that can shorten wait times and enhance customer satisfaction. This website suggests a web-based queue management system that enables users to get tickets using their smartphones, get alerts when their turns are about to happen, and rate and review their overall experience. To optimize their service delivery, service providers can also make educated decisions by using the system to track and evaluate customer behavior and queue status. The creation of an automated queue management system that can improve client satisfaction and service effectiveness is the goal of this design. Queue is a linear data structure which follows a particular order in which the operations are performed.

   The order may be **FIFO.**

II. Keywords

1. **Queue**: is a technique for setting up people or things that must be handled or served in a certain order. a list of commands, data items, etc. that are saved and can be retrieved in a specific order—typically the order of insertion.
2. **QUTAS**: Queue management System in University of Information and Applied Sacince.
3. **QMS***:* Queue management System
4. **(SLA):** Service Level Agreement a formal agreement that defines the expected level of service between a service provider and its customers.
5. **Integration:** Connecting the queue management system with other software to streamline data exchange and enhance the customer experience. Examples include customer relationship management (CRM), appointment scheduling, and point-of-sale (POS) systems.
6. **Queue Segmentation:** Dividing the queue into segments based on criteria such as service types, customer groups, or appointment types to streamline operations and optimize service delivery.
7. **FIFO**: First In First Out. It is type of the order in queue.

III. Innovative Idea/Concept

A website that connects clients with service providers in real time and enables them to make or cancel appointments online is an inventive concept for a queue management website. Additionally, the platform would offer accurate wait times, and queue status to customers. By lowering no-shows, raising customer satisfaction, and increasing service effectiveness, the platform would be advantageous to both customers and service providers.

IV. Introduction

The Department of Student Affairs and Registration is experiencing several issues, such as:

1. An inconsistent wait time in the department.
2. Difficulty in booking a waiting number
3. non-appearing transactions involving staff.
4. students face challenges in getting the right person to help them with their problem.
5. The system's accessibility to employees is concealed.

In this project for the Department of Student Affairs and Registration, we offered multiple solutions to address the issues facing the department:

1. We'll make use of the function that allows us to scan a barcode to get the waiting number.
2. We schedule the time that students must wait in the department to finish their query.
3. We provide the staff with information regarding the number of students on the waiting list and those who have finished their query.
4. We include a button to accept regular transactions on the staff page.
5. The student's name, the transaction he wishes to complete, and the staff members who will address his issue are displayed on the screen.
6. We display which staff members have accessed the system and processed transactions.

V. Objectives

General

The goal of our project is to create a system that will enable the Student Affairs and Registration Department to welcome students and assist them with resolving a variety of university -related issues, including modifying their specialization, obtaining graduation certificates, and altering their academic schedule, among many others. As a result, we made the decision to create an efficient system for the department's reception that will benefit both students and staff, thereby cutting down on the amount of time students must wait for their problems to be resolved. Additionally, we created a website for students where they can take a number and wait to have their issue resolved quickly.

Specific

The smaller, logically connected parts of the general objective can be broken down into the following:

- Allocating resources effectively is necessary to guarantee that employees are used effectively and that lines are managed effectively.
- Optimizing the queue management procedure to shorten wait times and raise customer satisfaction is known as queue optimization.
- Data analysis: To understand customer traffic patterns and make wise staffing and resource allocation decisions, data gathered by the queue management system is analyzed.[2]

The following are the results you hope to obtain from carrying out this project or research:

- Enhanced customer satisfaction: Your goal is to enhance customer satisfaction by optimizing the customer experience and cutting down on wait times.
- Enhanced efficiency is what you hope to achieve by streamlining the queue management procedure and wisely allocating resources.

VI. Scope and Limitation of the Project

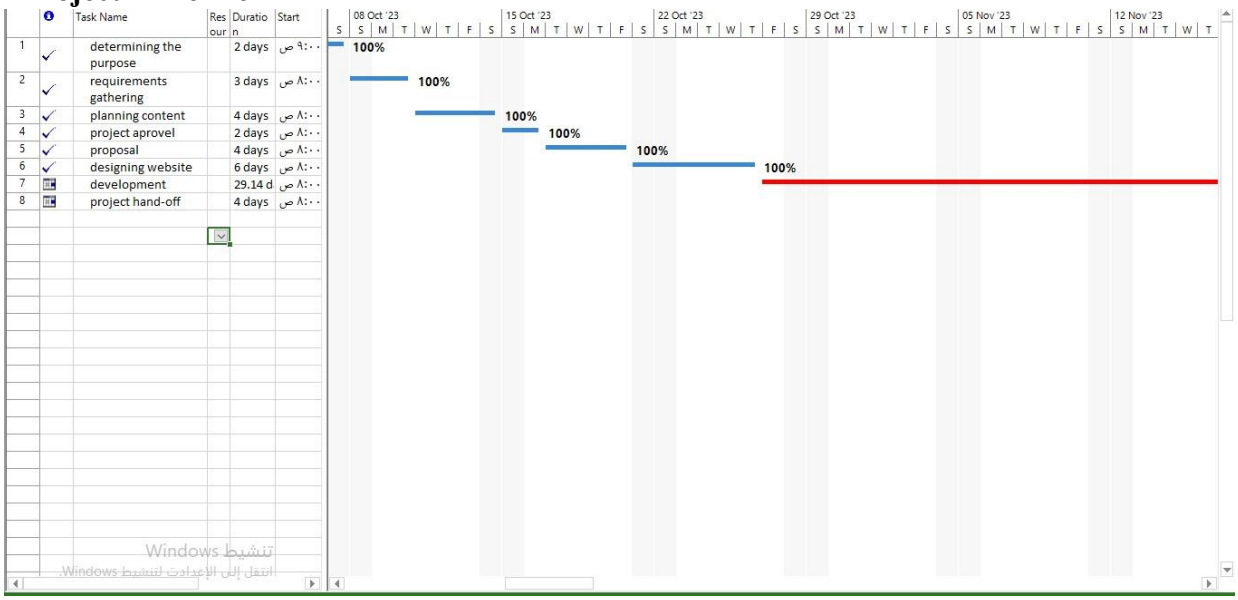
Creating a website for a queue management system is the extent of our project. And identifying which queue management system best meets the needs.

As a result, the project will entail investigating various queue management systems and their features.

The following are possible project limitations:

- Resources for hardware and software availability: The project might need hardware, such as a TV to display the order number, and software.
- Absence of prior research on the subject: The amount of prior research on queue management systems may be minimal, which may restrict the scope of our investigation.
- Methods, tools, and strategies employed in the data collection: The validity of our research may be compromised by inaccurate or unreliable data collection methods, tools, and procedures.
- Restricted access to data: The quantity and quality of data gathered may suffer from limited access to data.
- Time restrictions: Time restrictions may restrict the scope of the analysis and the quantity of research that can be done.

VII. Project Timeline



Review of Related Literature

II.1. Introduction

What did they use in the past?
What are the traditional methods of Queue Management?

- Physical Queue: In the past, the most common method of queue management involved physical queues, where customers would stand in line to wait for their turn. This method relied on manual monitoring and organization by staff members.

- Ticketing System: Another traditional approach was the use of ticketing systems. Customers would take a numbered ticket upon arrival and wait for their number to be called or displayed on a screen. This method helped maintain order and reduce confusion.

- Signage and Signboards: Signage and signboards were often used to guide customers and provide information about the queue. They would indicate where to line up, display instructions, or provide estimated waiting times.

- Queue Marshals: Queue marshals or attendants were employed to manage the queues, ensure proper order, and assist customers as needed. They would direct customers to the appropriate lines and address any issues or concerns.

History of Queue Management:
The history of queue management can be traced back to the early days of commerce and service provision. Here is a brief overview:

- Ancient Times: Queues have existed since ancient times, where people would wait in line for various services or goods. Examples include waiting for food distribution in ancient civilizations or waiting for entry into public events.

- Industrial Revolution: With the rise of industrialization and urbanization, queues became more prevalent. The growth of cities led to increased demand for goods and services, resulting in longer waiting times and the need for better queue management.

- Early Queue Management Systems: The development of queue management systems can be attributed to the need for efficient customer service. The first known implementation of a queuing system was in the 1970s at a post office in Sweden [1]. This system allowed customers to push a button for different services, creating multiple queues and prioritizing shorter transactions.

- Technological Advancements: Over time, queue management systems evolved with technological advancements. From simple black boxes with microchips to more

complex systems using PCs, the focus remained on improving customer flow and reducing wait times [2].

Traditional queue management methods face several challenges that can impact customer satisfaction and operational efficiency. Some of the key challenges include:

- Crowded Waiting Areas: Traditional queue management methods often result in crowded waiting areas, which can lead to customer dissatisfaction and negative impressions [3]. When customers are gathered in large crowds, it can create unrest and annoyance, potentially causing customers to switch to a different service provider. This challenge can also impact the reputation of the business in the market.

- Customer Experience: Customer experience is crucial in the service-based industry, and traditional queue management methods may not effectively handle customer flow, resulting in longer wait times and decreased customer satisfaction [4]. In competitive markets, such as Dubai, UAE, where customer expectations are high, banks need to be proactive in improving the customer experience to retain and attract customers.

- Wait Time: Long wait times can significantly impact customer satisfaction and may lead to customers considering switching to another service provider [4]. Traditional queue management methods often do not provide estimated wait times, causing customers to feel stressed and uncertain about how long they need to wait. Providing estimated wait times can reduce customer stress levels and improve their experience.

- Customer Journey: Designing an efficient and seamless customer journey is essential for success in the banking sector [5]. Traditional queue management methods may not effectively manage customer flow, resulting in a disjointed customer journey. Banks need to prioritize the convenience and ease of service delivery to ensure a positive customer experience.

- Business Intelligence: Traditional queue management methods may lack the ability to collect and analyze valuable business intelligence data [5]. Gathering statistical usage data and customer feedback is crucial for understanding customer experiences, measuring performance, and making strategic decisions for the future. Traditional methods may not provide the necessary tools and integration capabilities to collect and utilize this data effectively.

The long, overnight queue is seen as a miniature social system faced with the problems of every social System, formulating its own set of Informal rules to govern acts of pushing in and place keeping leaves of absence, and the application of sanctions. Cultural values of egali-Arianism and orderliness are related to respect for the principle of Service according to order of arrival which is embodied in the idea of a queue. The importance of time in Western culture is reflected in Rules relating to "serving time" to earn one's position in line, and to the regulation of "time-outs." The value of a business enterprise is expressed in the activities of professional speculators and queue "counters." Queue jumping is discouraged by a number of constraints, but, if social pressure fails, physical force is seldom used to eject the intruder. Principles of queue etiquette are illustrated with empirical and anecdotal evidence from the study of Australian football queues. This paper deals with the kinds of formal and informal arrangements made in queues to regulate behavior, to recognize the priority of early-comers, and to inhibit the growth of conflict. The study of queues is an attempt to describe the patterns

of behavior which attend waiting in lines where a great deal of time, discomfort, and risk of disappointment must be endured before the desired commodity is obtained. Our focus is on the forms of etiquette developed to regulate life in the queue and to minimize the amount of suffering experienced while waiting. The queue culture provides direction on such matters as place-keeping privileges, sanctions against pushing in, and rights of temporary absence from the waiting line. Evidence that the queue constitutes an embryonic social system is based primarily on the study of long, overnight lines for football tickets in Melbourne, Australia,

THE STUDY

Every Saturday afternoon in the month of September, over 100,000 spectators crowd into a stadium in Melbourne, Australia, to watch the "World Series" of Australian Rules football. On August 15, 1967, approximately 10,000 people formed twenty-two queues outside the Melbourne Football Stadium to buy 14,000 sets of tickets for the four games. It was the last opportunity to get tickets because 1 this study was supported by a grant from the Comparative International Program of the Department of Social Relations, Harvard University. The helpful suggestions of Thomas Cottle, George Humans, Alex Inkeles, Irving Janis, and Gerald Platt are gratefully acknowledged. I wish to express my gratitude to Ruth Chaplin and Frank Nordberg for their assistance in collecting and analyzing the data. Mail applications for the bulk of the tickets had been oversubscribed weeks before. A great many of the 10,000 faced disappointment because most queues usually buy the full allotment of two adults and two children's tickets.

From 6 A.M. until 8 A.M., when the selling windows opened, a team of nine research assistants, male psychology majors from the University of Mel-bourne, conducted short, standard interviews with 216 people in ten of the twenty-two queues. Each interviewer was randomly assigned to a queue.

Starting with the first person in line, the procedure was to approach every tenth person. The request was brief and informal: "I am from the University, and we are doing a study of how people feel about the queues. Would you care to answer a few questions?" Only two refusals were encountered; except for one queue, all interviews were completed by 8 A.M. when the lines began to move. Questions covered attitudes toward the system of queuing, evidence of pushing in and place keeping, arrangements to make the task of queuing more pleasant, as well as estimates of position in line and chances of getting a ticket. Interviewers also made notes on their observations of the physical shape of the queues and on their impressions of the mood and morale of the people in them. Data from the Melbourne stadium are the main source of evidence cited in this paper. Members of the research team also conducted interviews and made observations in the club queues at suburban Collingwood, Carlton, and Richmond, each of which had allocations of 1,000 tickets for club members. Data gathered from the club queues provide additional evidence presented in this paper.

THE QUEUE TRADITION

The system of selling seat bookings for the football finals several weeks before the start of the series was first introduced in 1956. Before 1956, people queued outside the stadium on the day of the game, and "first in" took the best seats. Over the years, because of the large increase in the number of football followers, the queue system became accepted as the only workable method for selling tickets. Although there were complaints from the public, the great overnight queues became a regular event at the end of a Mel-bourne winter. And as the queues took on an institutional character, increasing numbers of veterans began to regard them as a kind of cherished tradition or ritual. For example, even during the regular season, although it was possible to get a choice of seats two hours before the commencement of most Saturday games, long queues formed outside stadiums on Friday, perhaps to train for the big one in August. The queue of 1965 was perhaps the most remarkable, for in that year 25,000 people waited for

12,500 tickets, some of them for over a week, in mud and drizzling rain. Queues erected a shantytown of tents and caravans outside the stadium, and conditions, according to the Melbourne town clerk rapidly became "squalid and unhygienic." In 1966, to prevent a recurrence of the shantytown, the Melbourne City Council banned tents.

II.2. Research Review

a. Current State

What is the current State for QMS?

The current state of queue management systems in banks has evolved significantly from the traditional take-a-number model. Sophisticated tools and technologies are now available to optimize customer flow, enhance service efficiency, and improve customer experiences [6].

Challenges in Traditional Bank Queuing Systems:

- Long Waiting Times: Traditional queuing methods, where everyone waits in a single line, can lead to longer waiting times and poor customer experience [2].
- Customer Frustration and Dissatisfaction: Mismanagement and unorganized queuing contribute to high levels of customer dissatisfaction, leading them to reconsider their choice of bank [6].
- Inefficient Resource Allocation: Overcrowding and improper resource allocation can result in imbalanced workloads, longer waiting times, and discomfort for clients [6].
- Lack of Real-Time Insights: Traditional queuing systems lack advanced software to collect real-time data on customers, hindering data-driven decision-making and service improvements.

Benefits of Implementing Bank Queue Management System:

- Improved Customer Experience:
- Reduced Waiting Times: Queue management systems enable efficient customer flow, minimizing wait times and preventing frustration.
- Personalized Service: By collecting consumer data and analyzing patterns, banks can offer personalized attention and tailored offers to customers [7].
- Increased Customer Satisfaction: A queue management system significantly boosts overall customer satisfaction, leading to increased loyalty and positive word-of-mouth referrals [6].
- Enhanced Staff Productivity and Efficiency:
- Optimized Resource Allocation: Queue management systems allow banks to allocate staff resources effectively, reducing customer wait times during peak hours [8].
- Staff Performance Monitoring: Banks can track and evaluate staff performance metrics, identify areas for improvement, and provide targeted training to enhance efficiency [6].
- Streamlined Workflow: Automated queuing processes eliminate the need for manual ticket distribution, streamlining the workflow within the bank [7].

- Real-Time Analytics and Insights:
- Customer Behavior Analysis: Queue management systems provide valuable insights into customer preferences and patterns, enabling banks to allocate resources accordingly [5].
- Service Performance Evaluation: Data-driven metrics such as average wait times and service durations help assess service efficiency and optimize resource allocation [3].
- Data-Driven Decision Making: Customer feedback data allows banks to make informed decisions, adjust staffing schedules, and introduce self-service options to expedite processes.
- Features to Look for in a Queue Management System for Banking:

- Integration Capabilities: Seamless integration with existing banking systems, customer databases, and digital signage solutions.
- Multiple Service Channels: Support for in-person counters, self-service kiosks, and virtual queues accessible through mobile apps or online platforms.
- Appointment Scheduling: Functionality to allow customers to book specific time slots for their banking needs, ensuring efficient use of resources and minimizing waiting time [3].
- Mobile Applications: Mobile apps that enable customers to remotely join the queue, receive real-time updates, and manage their banking appointments.

Queue Monitoring and Reporting: Comprehensive features for bank staff to monitor customer flow, service times, and identify bottlenecks in real-time.
Integration with Customer Relationship Management (CRM) Systems: Access to customer profiles, transaction histories, and personalized preferences to deliver a more personalized customer experience.
The current state of queue management systems is characterized by significant growth and advancements driven by technological innovations and changing customer expectations. Here are some key trends and insights:

- Market Growth: The queue management system market was valued at US$ 482.73 million in 2022 and is expected to grow at a CAGR of 5.5% during 2022-2030 [1].

- Virtual Queuing: Virtual queuing systems have become more prevalent, allowing customers to join a queue remotely through a smartphone app and receive real-time updates [2]. This approach reduces physical waiting times and overcrowding in waiting areas.

- Mobile Tickets: Businesses are adopting mobile ticketing solutions, where customers can sign in using their smartphones and receive updates and wait times through SMS messages and push notifications. This streamlines the check-in process and frees up staff to focus on customer care.

- Video Consultations: Video appointment software enables businesses to offer remote consultations, providing convenience and flexibility for customers. This trend emerged during the pandemic and is likely to continue as customers expect more flexible appointment options.

- Real-time Insights: Queue management systems are being integrated with other data sources to provide real-time insights into the customer journey [3]. This integration improves operational efficiency and enhances the overall customer experience.

- AI and IoT Integration: Artificial Intelligence (AI) is being leveraged to optimize queue management systems and improve the customer experience. AI-based systems analyze real-time customer data to optimize customer flow and meet key performance indicators. Additionally, the Internet of Things (IoT) enables seamless communication between devices, allowing for improved customer flow management and staff scheduling.

- Intelligent Queue Management: Organizations are seeking more customized and industry-specific queue management systems. AI-driven systems can adapt to specific business requirements and continuously learn and improve based on gathered data and insights.

- Rapid Deployment: Queue management systems are now faster to deploy, allowing organizations to implement solutions quickly. This was particularly important during the pandemic when businesses needed to adapt to social distancing regulations.

b. Research Gaps

Why are we working on this idea when there are so many queue management websites?

These days, there are several web applications regarding queue management systems, but they are often poorly designed and have numerous flaws. Also, they fail to place tasks at the best possible machine, since they lack cluster-wide information, leading to worse job completion time, especially for heterogeneous workloads.[7]
Managing the queue is essential to providing excellent customer service. How you handle the line will determine how a consumer feels about your company from the first.

Here are a few instances of inadequate queue management:
1. There is nothing in the queue:
Customers will become lost and irritated if you don't have a structure set up to help them through the procedure. Nobody enjoys having to wait around for a response or a product. More complicated problems like "abandoned" shopping carts and unfinished orders may also result from this.

3. User Interface and Experience: Customers, staff members, and managers find it challenging to use the user interface. Furthermore, there are several issues with

authorization and licensing. For example,
when a consumer visits the website pretending to be a manager or employee, this leads to a lot of issues and unhappy customers. It can be difficult to create an interface that is both straightforward and user-friendly while also considering the various demands and preferences of users. A confusing or complex user interface might make it harder for users to understand and utilize the system.

4. Ineffective organization and incorrect information delivery: Many websites have stunning designs, but they don't properly offer the necessary information.

5.Scalability: The system's capacity to manage heavy traffic and lineups may become problematic as the number of users rises. It is essential to make sure the system can be developed to handle an increasing number of users.

6.Performance: User annoyance may arise from sluggish interfaces, slow loading times, or delays in updating queue statuses. It is crucial to optimize system speed to offer a seamless and effective user experience.

7.User Adoption: It might be difficult to persuade users to accept and make use of the queue management system. User adoption may be hampered by reluctance to change, ignorance, or a poor comprehension of the system's advantages.

8.Integration with Current Systems: It might be difficult to integrate the queue management system with other current systems, including customer databases or appointment scheduling software. Planning carefully and taking technological concerns into account may be necessary to ensure smooth data flow and prevent data duplication or inconsistencies.

9.Mobile Responsiveness: It is imperative that the queue management website be completely adapted for mobile platforms, given the growing prevalence of mobile devices. It ought to provide a smooth and intuitive experience on a range of screen sizes and operating systems.

10.Security and Privacy: It's critical to safeguard user information and make sure the queue management system is secure. Important things to think about include putting strong security measures in place, encrypting data, controlling access, and following privacy laws.

11.System Reliability: System failures or downtime can cause operations to be disrupted and the user experience to be severely impacted. To guarantee system availability, strong and dependable system architecture, backup and recovery methods, and routine maintenance are essential.

12.Customization and Flexibility: The queue management system must consider the needs and processes of each organization. It might be difficult to offer customization choices and adaptability to various organizational structures and procedures.

13.User Support and Training: It is essential to provide users with sufficient training materials and user support resources to enable them to comprehend and make efficient

use of the system. Insufficient guidance or assistance might impede user adoption and result in discontent among users.

c. Motivation

Motivation plays a crucial role in queue management systems as it helps to structure and manage waiting lines effectively. Understanding the factors that motivate people to join queues can help businesses and organizations optimize their queue management strategies. Here are some motivations commonly observed in queue management systems:

1. Desire for Cheap Products: When businesses offer discounted or low-priced products, it can attract a large number of people willing to wait in queues to avail of these deals.

2. Fresh Start-Ups: The launch of new and credible initiatives, coupled with strategic marketing, can create a magnet for large crowds of people. For example, the opening of the first McDonald's in Moscow triggered long queues of people eager to experience the brand

3. Launch of Popular Items: The release of highly anticipated products such as video games or critically acclaimed movies can lead to a huge influx of people queuing up to get their hands on these items. For instance, the waiting lines for popular video games like GTA V are a testament to the motivation of fans to acquire their copies.

4. Celebrity Autographs: Fans often form long queues to get autographs from their favorite celebrities. This motivation is commonly observed in the entertainment industry, where fans eagerly wait for the opportunity to meet their beloved stars.

5. Tourist Spots: Popular tourist destinations, such as Atlantis or Disney Land, often have long queues due to the unique experiences they offer. Despite the waiting time, people are motivated to bear the queues to enjoy the attractions these locations provide.

6. Mouth-Watering Temptations: The allure of delicious food can motivate people to join queues, regardless of the distance or waiting time. For example, a hot dog vendor on the corner of a street can attract an infinite queue of people craving the mouth-watering delicacy [9]

7.Personal Needs: Queues for personal needs, such as gender-specific queues or toilet queues, are found worldwide. These queues serve specific purposes and are motivated by individual requirements.

8. Social Responsibilities: People often join queues for social causes, such as queuing up to donate blood. This motivation stems from a sense of selflessness and the desire to help others.

9. Disaster-Incited Provisions: In times of crisis or disaster, queues may form for essential life necessities, such as food or water. The motivation here is driven by the need to fulfill basic survival needs [9].

Understanding these motivations can help organizations design effective queue management systems that minimize waiting times and enhance customer satisfaction. Advanced technologies like Queue Management Systems and mobile apps are being adopted to streamline the queuing process and improve overall efficiency.[9]

d. Proposed Solution

What will we do to choose us?

Designing User Interfaces:
To comprehend user requirements and preferences, do usability testing and user research.
Create an interface that is simple to use and intuitive, with obvious navigation and visual signals.
Use responsive design strategies to make sure that the user experience is the same on all devices.

Privacy and Security:
Put robust authorization and authentication procedures in place.
Update and patch the system frequently to fix security flaws.
Respect applicable privacy rules and data protection laws.

Training and Support for Users:

- Create thorough user manuals, frequently asked questions, and online support materials.
- Offer customer service methods that are responsive, such live chat and email.
- Provide webinars, video tutorials, or training sessions to inform users about recommended practices and system usage.

TV in the waiting area:

- Real-time waitlist updates.
- Direct clients to the help desk.
- Use a customized strategy.
- Construct automatic alerts.

Analytics for services:

- Track and measure KPIs.
- Recognize the purpose of the customer's visit.
- Track the group's performance.
- Cut back on running expenses.

Visit planner:

- Present up-to-date location-related data.
- Permit clients to select the ideal time to visit.
- quieter waiting areas with sign-in from a distance.
- Aid guests in getting ready for their visit.

Reduce scheduling stress for faculty, staff, and students. QUTAS unlocks easy-to-use online appointment scheduling and virtual waitlists for any campus office.

Chapter III - REQUIREMENTS SPECIFICATIONS AND DESIGN

III.1. Introduction

An overview of the requirements, specifications, and design for your web application for the queue management system is given in this chapter. It provides an overview of the functional and non-functional requirements, the system's goal, and the system design methodology. The features and functionalities that your web application for the queue management system must have been specified in the functional requirements. The functions, operations, and behaviors that the system must be able to carry out are described in these requirements. Instead of focusing on functions, the non-functional criteria outline the attributes and traits your web application for the queue management system should have. The web application for your queue management system is designed during the system design phase. Your queue management system web application's unique requirements should be included in the system design to make sure it offers both administrators and customers an effective and user-friendly experience. We can create a reliable and efficient web application for our queue management system that satisfies user demands and enhances the queue management procedure by adhering to a methodical approach to requirements specification and system design.

The Department of Student Affairs and Registration at the University of Technology and Applied Sciences Ibra is a critical hub for academic administration, overseeing the enrollment and registration processes that are fundamental to the student experience. Unfortunately, the department is confronted with a series of challenges, notably exacerbated during registration days. These challenges not only strain the dedicated staff but also result in frustrating delays for the clients, comprising students and their families.

In recognizing the need for a comprehensive solution, the university has embarked on the development of a Requirements Specifications and Design initiative. This strategic endeavor aims to thoroughly analyze and document the existing issues within the registration process, understand the root causes of delays, and subsequently design a system that not only alleviates strain on the staff but also enhances the overall experience for clients.

This document will delve into the specific requirements that need to be addressed to optimize the registration process. By clearly defining these requirements, the university aims to establish a solid foundation for the subsequent design phase. Collaborative effort between stakeholders, staff, and technology experts will be crucial in crafting a solution that not only resolves current challenges but also anticipates and accommodates future needs as the university continues to evolve.

As we are getting started on this Requirements Specifications and Design journey, the overarching goal is to create a registration system that is not only efficient and responsive but also aligns seamlessly with the university's commitment to delivering a superior education experience to its students. Through a meticulous analysis of requirements and a thoughtful design process, the university endeavors to transform the Department of Student Affairs and Registration into a model of operational excellence.

III.2. Requirements

a. **Functional Requirements**
b. **Non-functional Requirements**

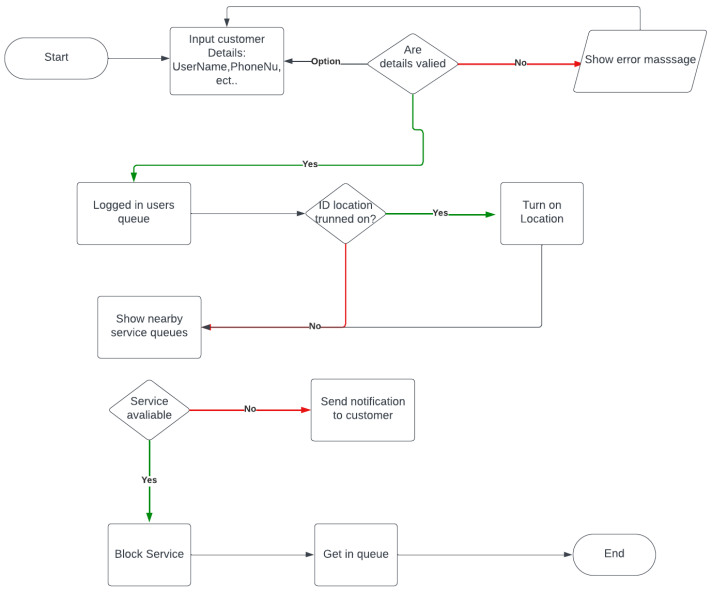
Functional Requirements	Non-functional Requirements
Online Registration System: Implement an online registration system that allows students to register for courses, update personal information, and make payments remotely. The system should be user-friendly, secure, and accessible from any device with an internet connection.	Performance: The registration systems should be highly responsive, capable of handling many concurrent users without significant delays or system crashes. The response time for online transactions and document verification should be minimal to provide a seamless experience for students.
Queue Management System: Introduce a queue management system that enables students to take virtual tickets and receive real-time updates on their position in the queue. The system should provide estimated waiting times and notify students when it's their turn to approach the registration counter.	Security: Ensure that all registration systems and databases adhere to robust security standards to protect student data and confidential information. Implement encryption protocols, user authentication mechanisms, and regular security audits to prevent unauthorized access and data breaches.
Automated Document Verification: Develop an automated document verification system that can scan and verify student documents, such as identification cards, transcripts, and certificates. This system should reduce manual effort, minimize errors, and expedite the document verification process.	Scalability: Design the registration systems with scalability in mind to accommodate future growth in student enrollment. The systems should be able to handle increasing demands without compromising performance or user experience.

Self-Service Kiosks: Install self-service kiosks at strategic locations within the department to allow students to perform routine tasks, such as printing registration forms, requesting transcripts, or updating contact information. These kiosks should be intuitive, reliable, and available during operational hours.	Accessibility: Ensure that all registration systems, online platforms, and physical facilities are accessible to individuals with disabilities. Adhere to accessibility guidelines and provide alternative methods of registration or assistance for students with special needs.
Staff Training: Conduct regular training sessions for department staff to ensure they are proficient in using new registration systems, handling technical issues, and providing effective customer service. Training should focus on improving efficiency, minimizing errors, and addressing common student queries.	Reliability: The registration systems should be reliable and available during scheduled registration periods. Implement backup and disaster recovery mechanisms to minimize downtime and ensure data integrity.
Mobile Application: Create a mobile application that provides students with access to essential registration services, notifications, and updates. The application should be compatible with different mobile platforms and offer a seamless user experience.	User Experience: Prioritize user experience in the design of registration systems and interfaces. Systems should be intuitive, easy to navigate, and provide clear instructions to minimize user errors and frustration.
	Integration: Ensure seamless integration between different registration systems, databases, and third-party applications. This includes integration with the university's student information system, financial systems, and academic platforms, enabling efficient data exchange and synchronization.

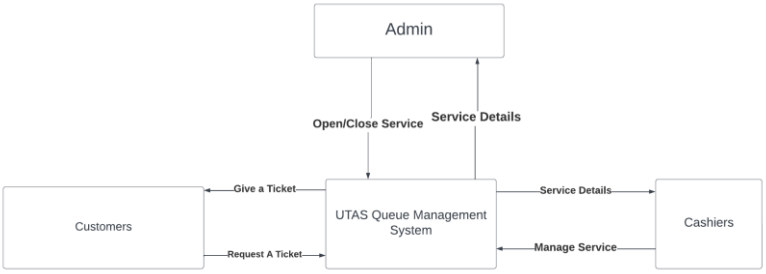
	Compliance: Ensure compliance with relevant regulations, data protection laws, and institutional policies. The registration systems should adhere to standards for data privacy, consent, and information security.
	Support and Maintenance: Provide ongoing technical support and maintenance for the registration systems, including timely bug fixes, updates, and enhancements. Establish a dedicated helpdesk or support team to assist students and staff in case of system issues or inquiries.

III.3. System Design

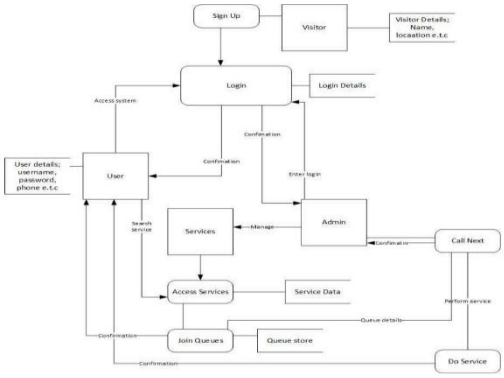
- **For Software/Application**
  - **Structured Systems Analysis and Design**
    - **Flowchart (Process)**



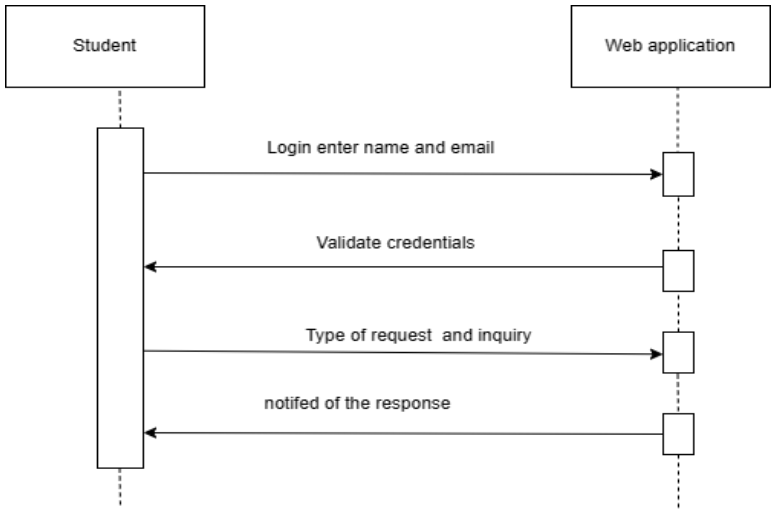
- **Data-flow Diagram (Data flow)**



- **Entity-Relationship Diagram (Data Design)**



- **Object-Oriented Analysis and Design**
  - **Entity-Relationship Diagram (Data Design)**
  - **Structure**
    - **Class Diagram**
  - **Behavioral**
    - **Use Case Diagram**
    - **Sequence Diagram**
    - **State Diagram**



Chapter IV - Project Development and Implementation

IV.1. Introduction

The actual creation and execution of the web application for the queue management system come under more scrutiny. An overview of the development process is given in this chapter, which also highlights important factors and procedures that must be followed to implement the system. It establishes the framework for the parts that follow, which go into detail on the prototype and the project's general progress. It outlines the aims, objectives, and important factors of this phase and provides context for the sections that follow.

1. Development Process: The steps involved in developing the web application for the queue management system are outlined in the Introduction section. It might discuss the selected software development methodology such as Waterfall or Agile and give a succinct rundown of the organization and procedures that will be followed throughout the process. It might also include a description of the duties and responsibilities of the project's development team members.
2. Important Things to Think About: The main factors that will direct the creation and execution of the web application for the queue management system are covered in the Introduction section. These

factors could include interoperability with various platforms and devices, security, performance, and scalability. Any legal or regulatory requirements that must be met during the development process might also be mentioned.

3. Stakeholder interaction: Throughout the planning and implementation stages, stakeholder interaction is crucial, as the Introduction section highlights. It emphasizes how important it is to have efficient lines of communication, work together, and get input from all relevant parties, including clients, end users, sponsors of the project, and others. It might refer to the channels and procedures that will be used to guarantee active stakeholder participation.

4. Project Scope and Limitations: The Introduction section provides a brief overview of the project's scope, detailing the features and functionalities that will be included in the web application for the queue management system. It might also address any restrictions or limitations that must be considered while developing a project, like resource or technological limitations or business needs.

The development and implementation phase of the web application for the queue management system is outlined in the Introduction part of this document. A high-level summary of the development process, goals, important factors, stakeholder involvement, and project scope is given. To ensure a seamless transition into the actual development and implementation activities, this part aids development teams and stakeholders in understanding the context and goal of the sections that follow.

IV.1. Prototype Description:

The early iteration of the web application for the queue management system is shown by the prototype. It's a working model that highlights the main characteristics and capabilities of the finished product. A thorough description of the prototype, including its purpose, design, and operation, can be found in the description.

1. Design of the User Interface (UI): The prototype displays the UI's visual components and layout. It displays the online application's overall design, typography, color palettes, and layout. The goal of UI design is to produce a user interface that is easy to use and aesthetically pleasing, improving user experience, and enabling smooth system interaction.

2. Main Features: The main goal of the prototype is to demonstrate the main features of the web application for the queue management system. Features including user registration, queue generation, customer addition, and real-time queue status updates are demonstrated practically. Making ensuring the prototype faithfully reflects the expected functionality of the finished system is the aim.

3. The queue management system web application's basic flow of activities and procedures is depicted in the prototype. It shows users how to use multiple screens, interact with different features, and carry out actions including joining queues, getting notifications, and keeping track of their queue position. The prototype's system flow aids in the understanding and visualization of the user journey by developers and stakeholders.

4. Sample Data and Simulations: The prototype may include sample data and simulations to produce a realistic experience. This enables people to engage with the system through simulated scenarios and watch how it behaves. For example, the prototype may provide sample notifications, mimic customer additions to a queue, or display the real-time queue status update. Without relying on actual live data, sample data and simulations offer an insight into the behavior and performance of the system.

Project managers, clients, and development teams can all see a physical depiction of the queue management system web application in the prototype description. It facilitates the process of obtaining input, pinpointing possible enhancements, and verifying the system's blueprint prior to entering the comprehensive development stage.

IV.1. Prototype/Model

- **For Software/Application**
  - **Website, Mobile Phone App, Desktop Applications, Hybrid/Cross-platform *(At least 50% for Diploma 2nd Year, Required for Bachelor)***
  - **Physical Database *(Required)***

DATABASE DESIGN

Database Name: qms

Tables:

1- Users Table:

#	الاسم	النوع	التجميع	الخواص	خالي	إفتراضي	التعليقات	إضافي	العملية
1	**ID**	int			لا	لا شيء		AUTO_INCREMENT	تغيير سقّط
2	**Username**	varchar(255)	utf8mb4_0900_ai_ci		نعم	*NULL*			تغيير سقّط
3	**PhoneNumber**	varchar(255)	utf8mb4_0900_ai_ci		نعم	*NULL*			تغيير سقّط
4	**Email**	varchar(50)	utf8mb4_0900_ai_ci		نعم	*NULL*			تغيير سقّط
5	**Problem**	enum('Problem 1', 'Problem 2', 'Problem 3')	utf8mb4_0900_ai_ci		نعم	*NULL*			تغيير سقّط
6	**created_date**	timestamp			نعم	CURRENT_TIMESTAMP		DEFAULT_GENERATED	تغيير سقّط
7	**status**	varchar(50)	utf8mb4_0900_ai_ci		نعم	waiting			تغيير سقّط

2- Admin Table:

#	الاسم	النوع	التجميع	الخواص	خالي	إفتراضي	التعليقات	إضافي	العملية
1	id	int			لا	لا شيء		AUTO_INCREMENT	تغيير سقّط أكثر
2	username	varchar(50)	utf8mb4_0900_ai_ci		لا	لا شيء			تغيير سقّط أكثر
3	password	varchar(255)	utf8mb4_0900_ai_ci		لا	لا شيء			تغيير سقّط أكثر

3- Cashiers Table:

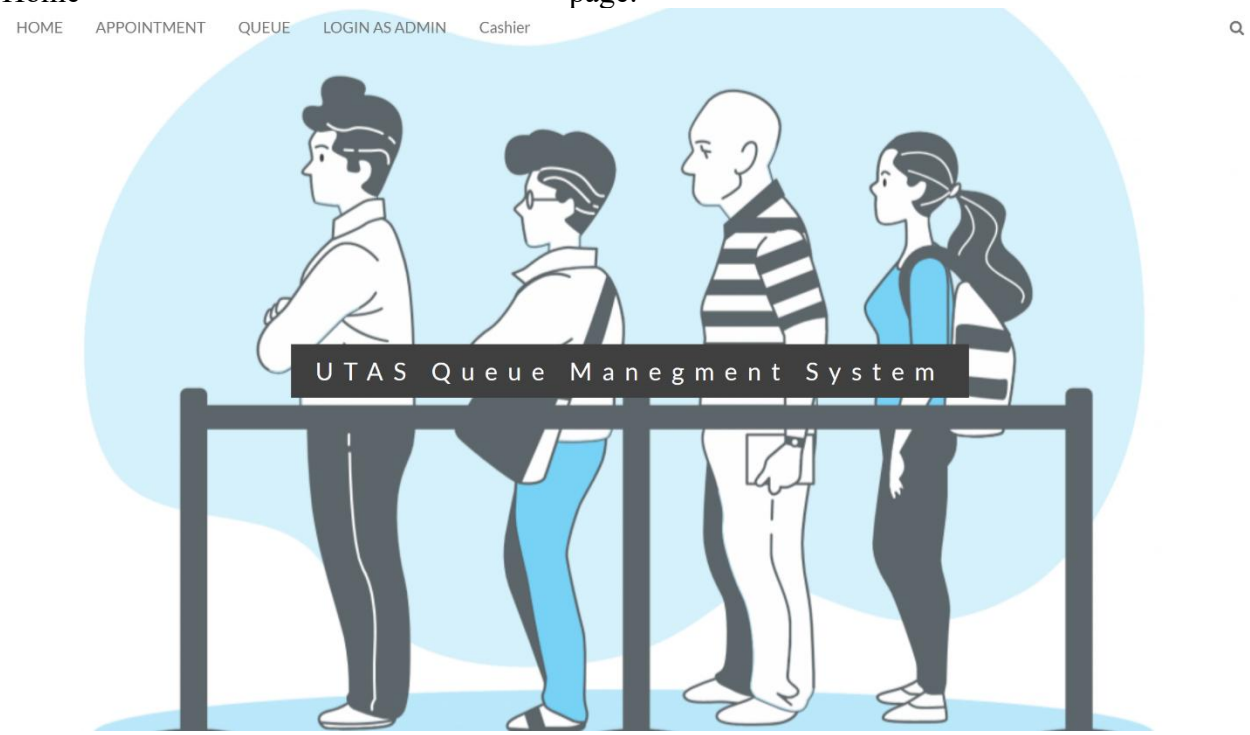
#	الاسم	النوع	التجميع	الخواص	خالي	إفتراضي	التعليقات	إضافي
1	id	int			لا	لا شيء		AUTO_INCREMENT
2	name	varchar(255)	utf8mb4_0900_ai_ci		لا	لا شيء		
3	problem	varchar(255)	utf8mb4_0900_ai_ci		نعم	NULL		
4	status	varchar(50)	utf8mb4_0900_ai_ci		لا	لا شيء		

4- Queue Table:

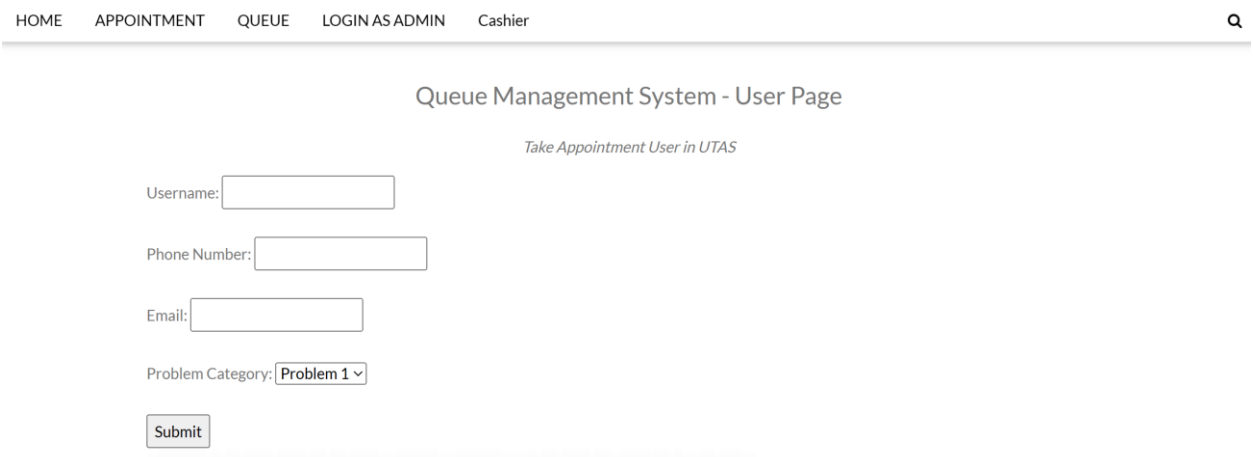
#	الاسم	النوع	التجميع	الخواص	خالي	إفتراضي	التعليقات	إضافي
1	id	int			لا	لا شيء		AUTO_INCREMENT
2	customer_name	varchar(100)	utf8mb4_0900_ai_ci		نعم	NULL		
3	phone_number	varchar(20)	utf8mb4_0900_ai_ci		نعم	NULL		
4	queue_number	int			نعم	NULL		
5	current_serving_number	int			نعم	NULL		
6	created_at	timestamp			نعم	CURRENT_TIMESTAMP		DEFAULT_GENERATED
7	current_number	int			نعم	0		

INPUT DESIGN

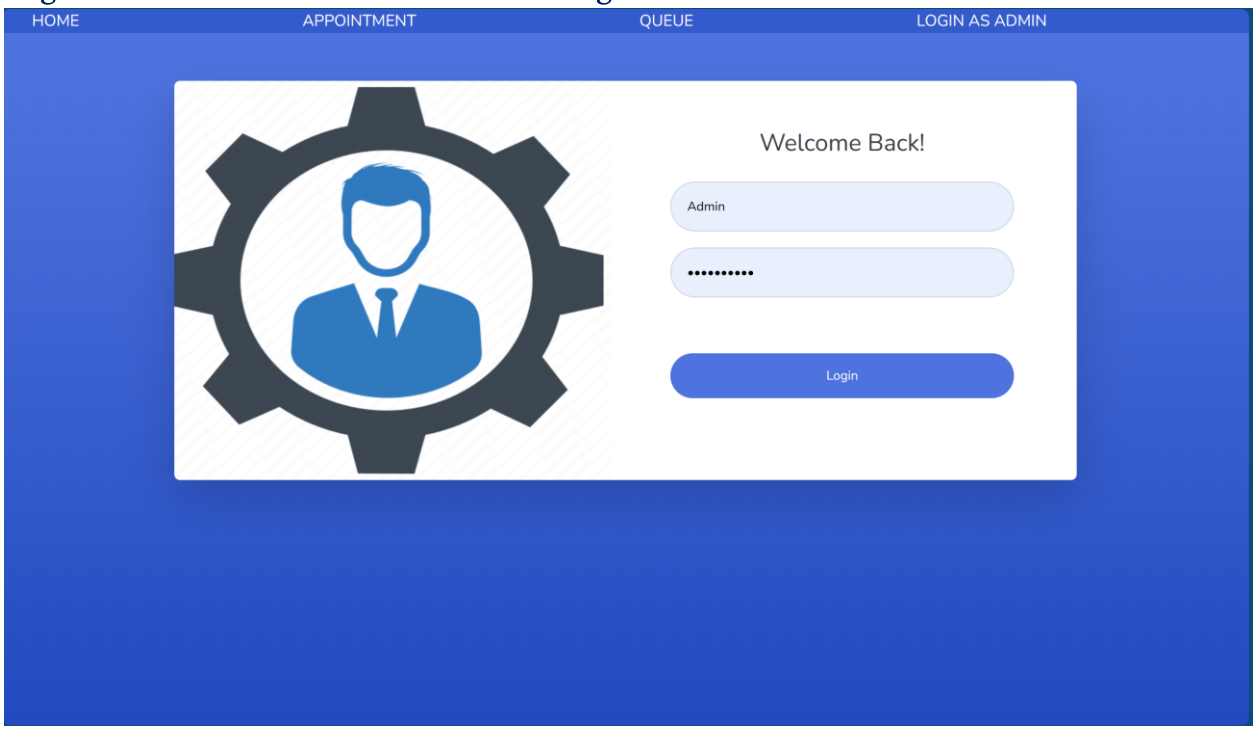
Home page:



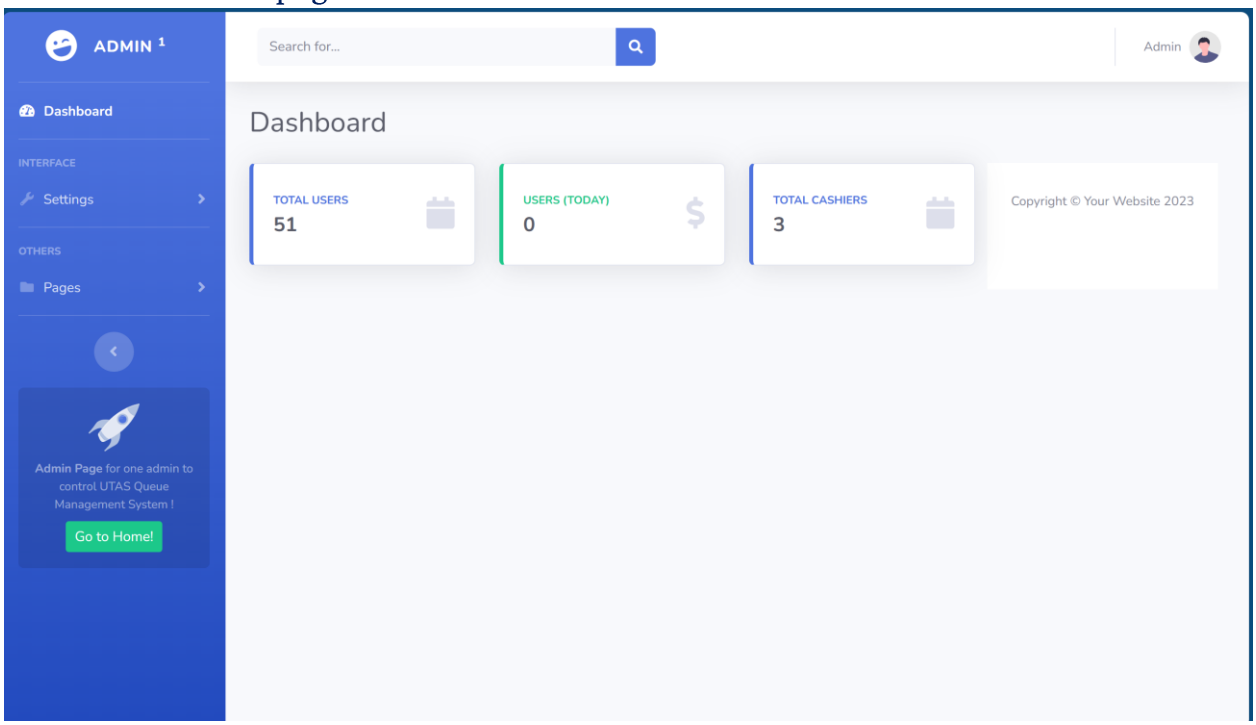
Appointment page:



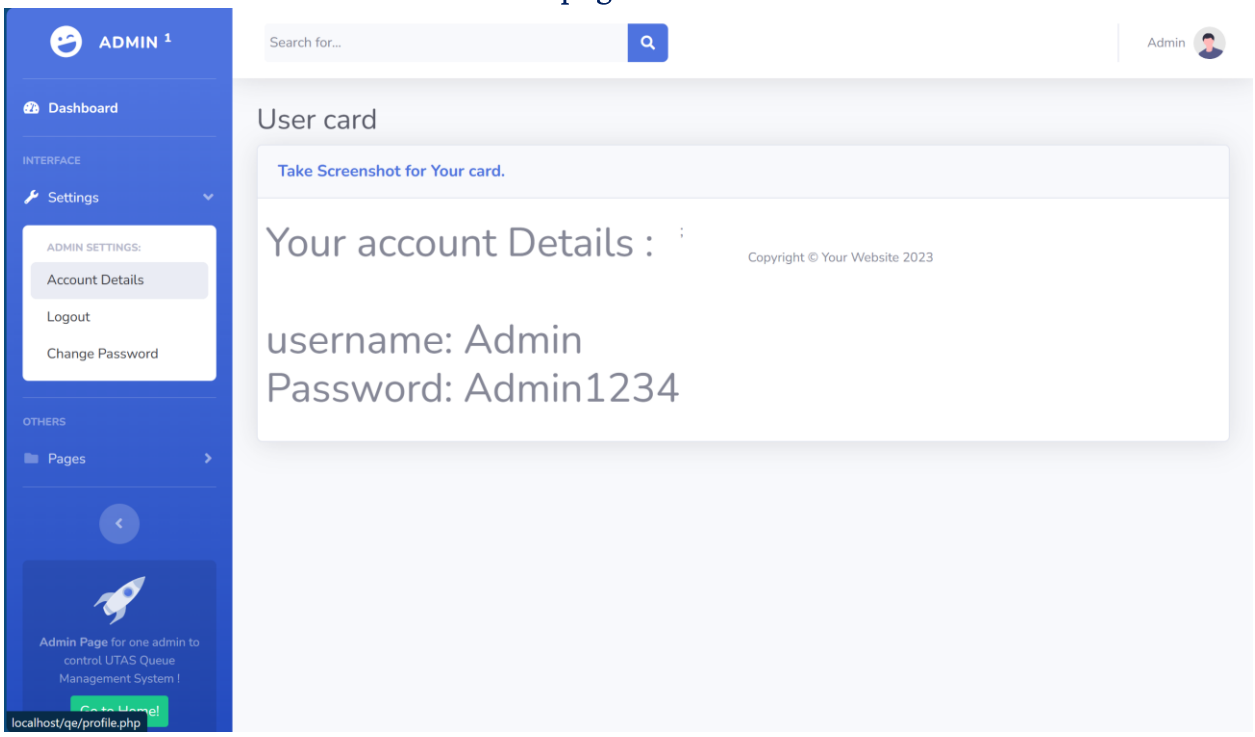
Login As Admin Page:



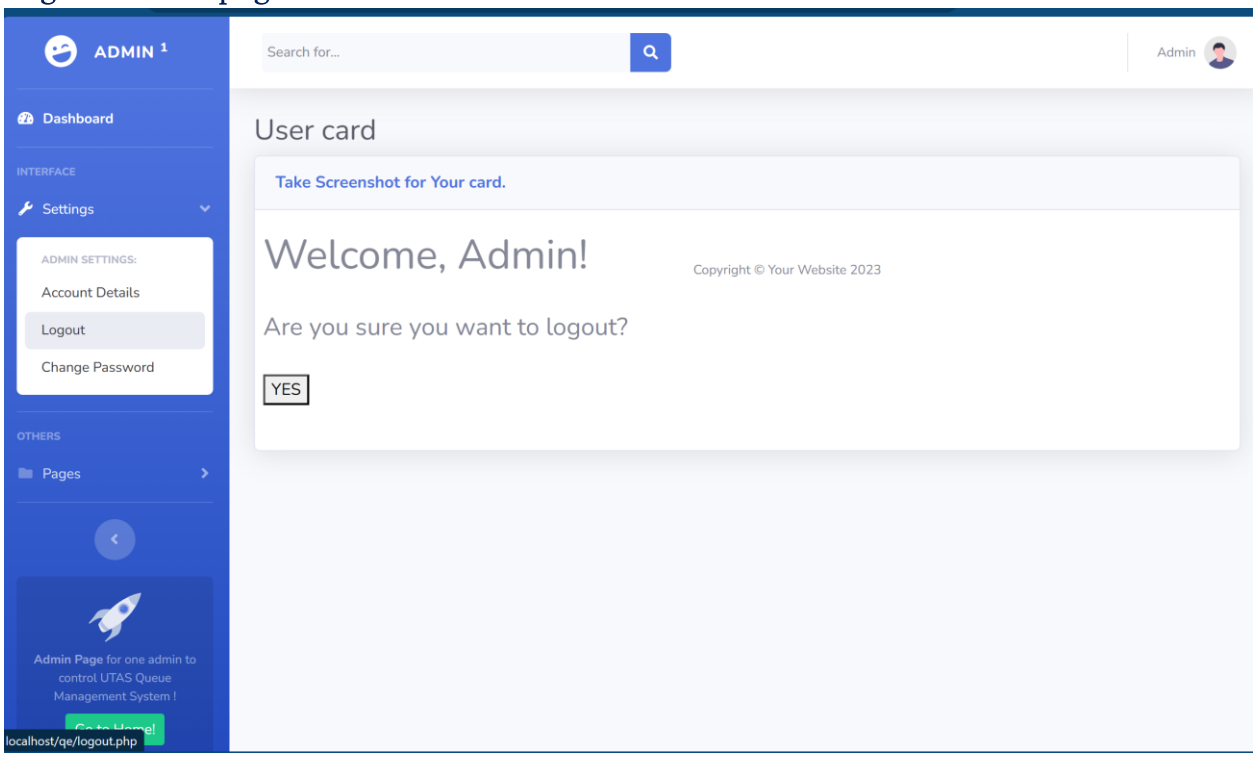
Dashboard admin page:



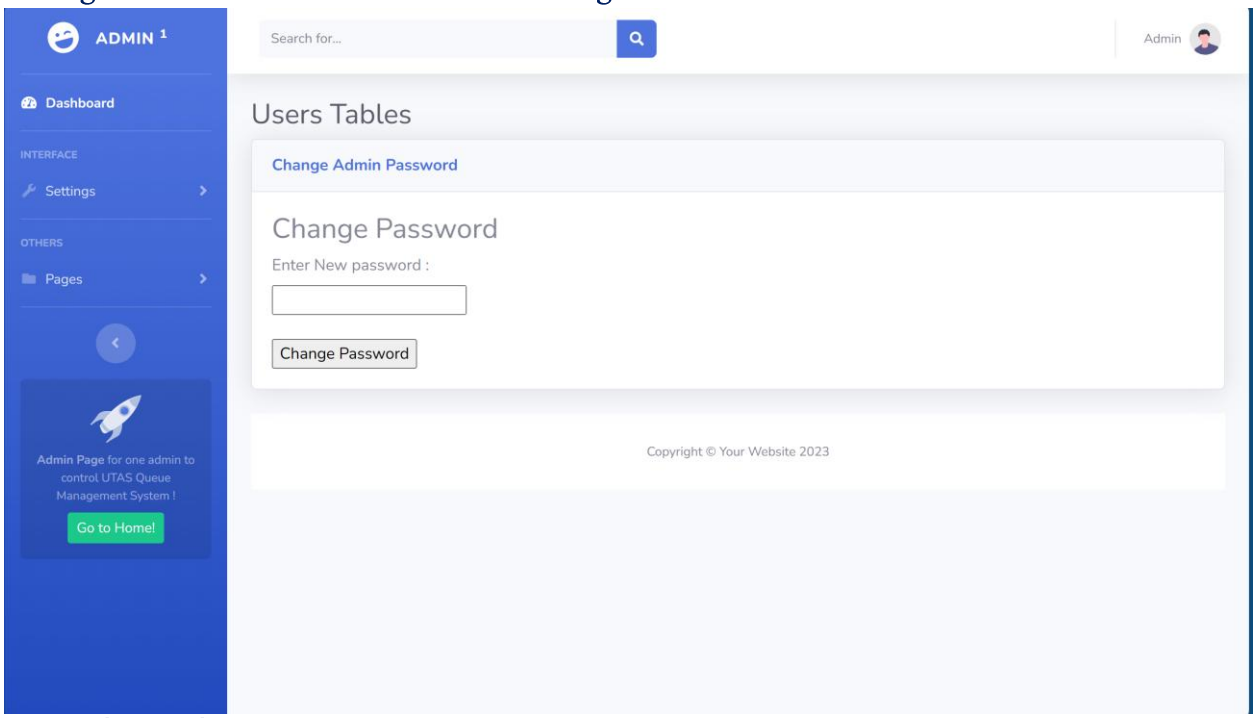
Admin Profile page:



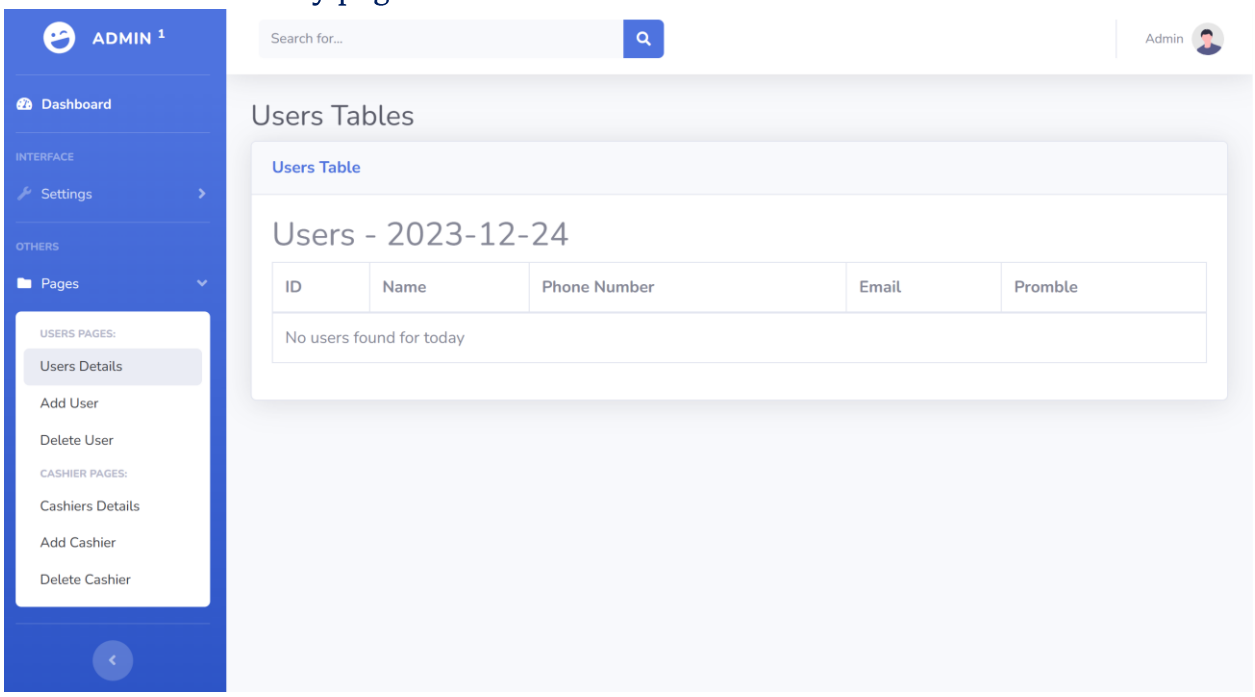
Logout admin page:



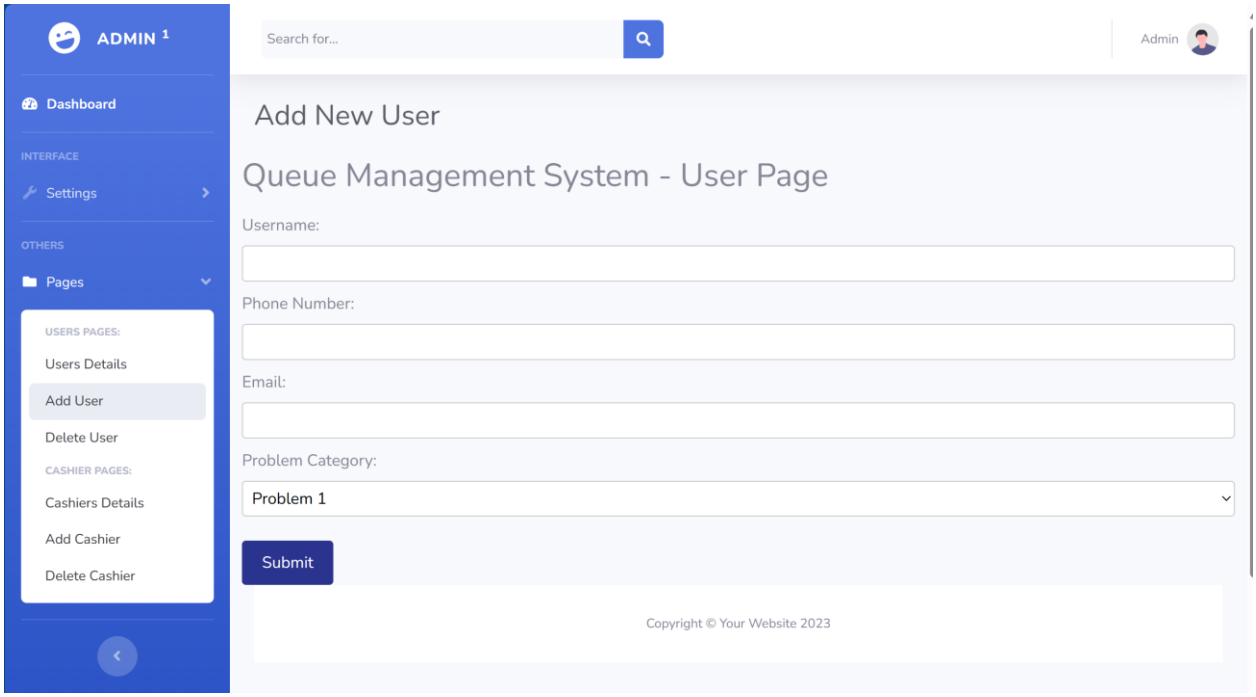
Change Password Admin Page:



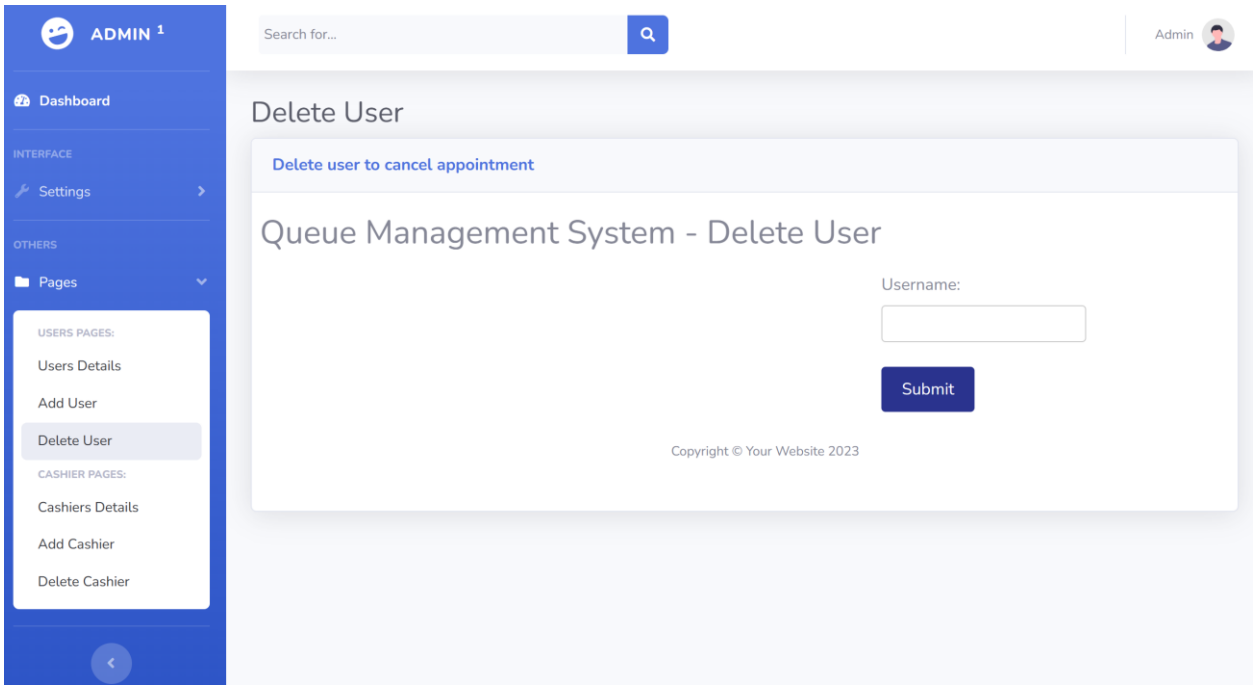
Users found for today page:



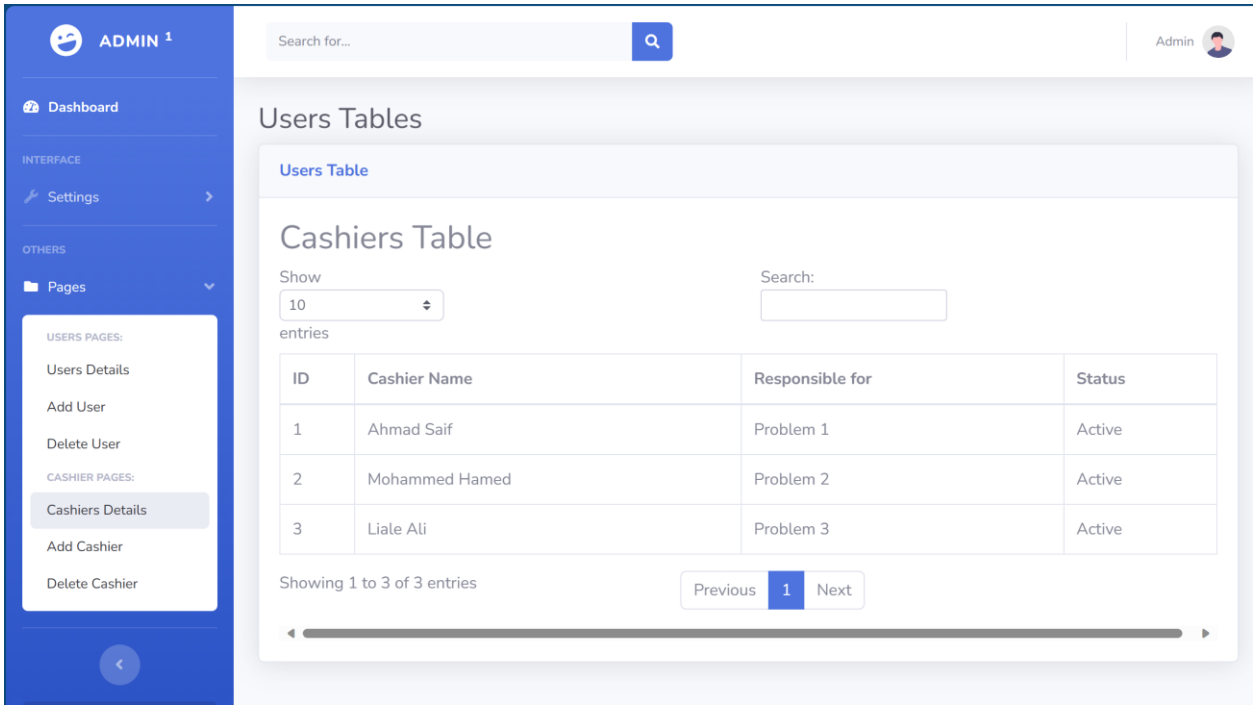
Admin can Add New User:



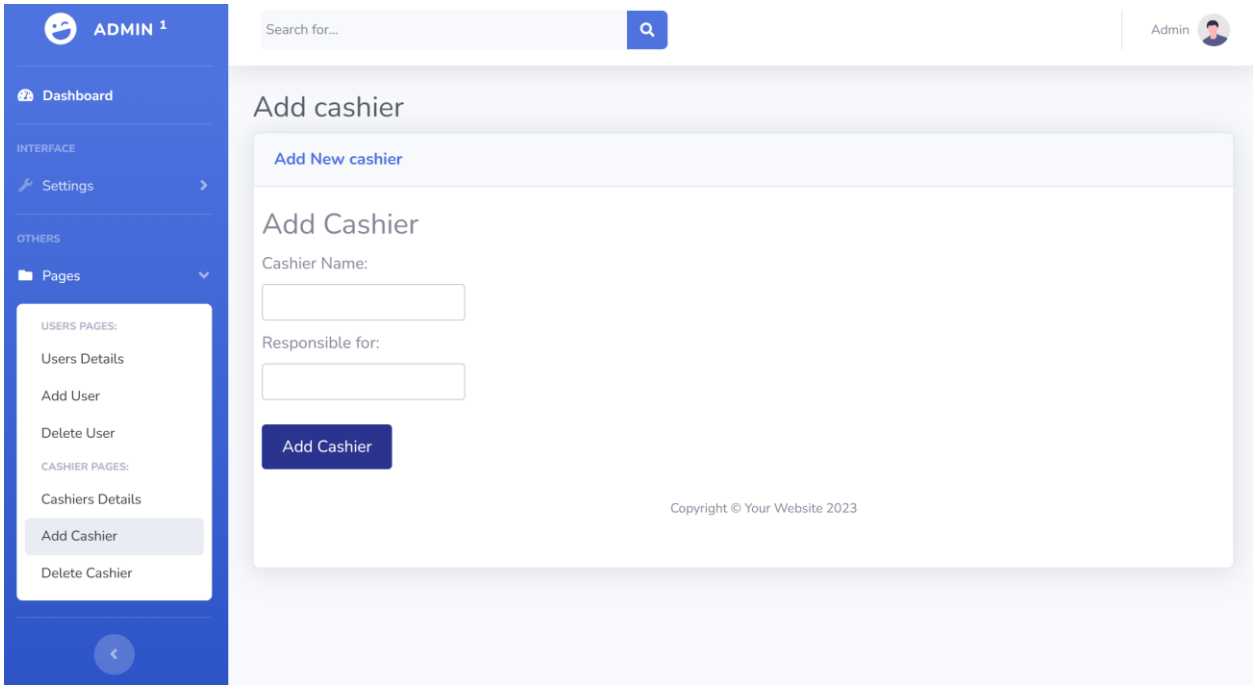
Admin can Delete User:



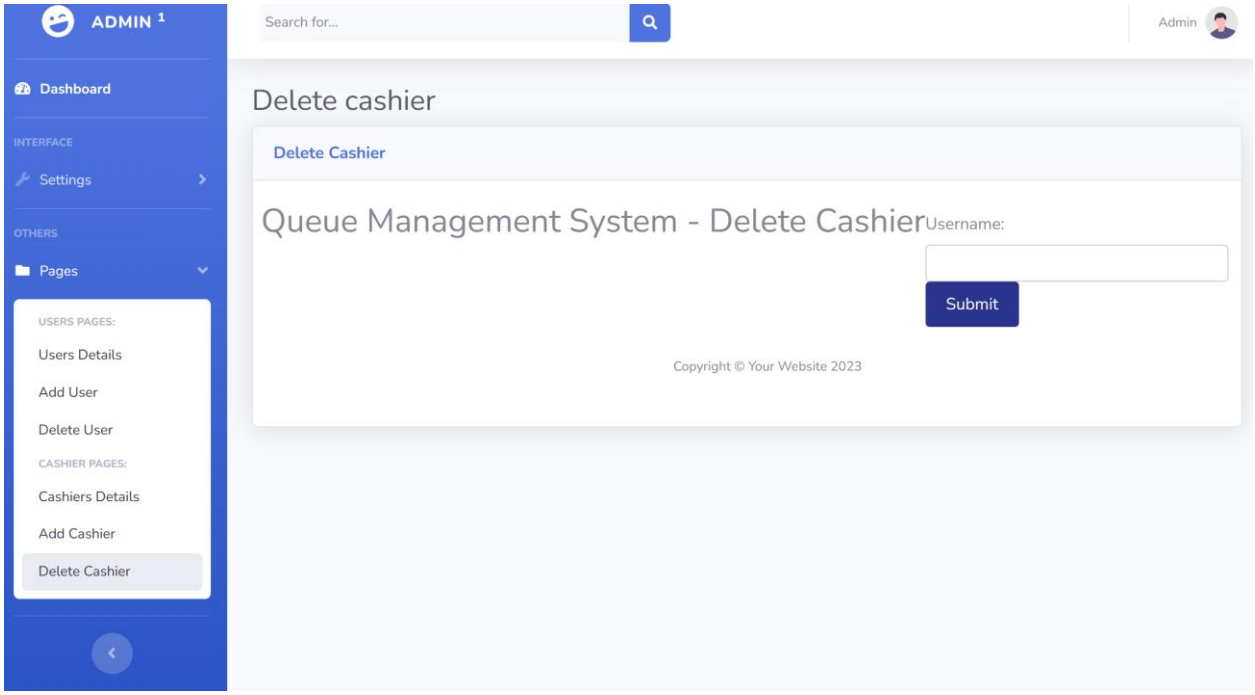
Cashiers Details:


ADMIN 1
Search for...
Admin
Dashboard
INTERFACE
Settings
OTHERS
Pages
USERS PAGES:
Users Details
Add User
Delete User
CASHIER PAGES:
Cashiers Details
Add Cashier
Delete Cashier
Users Tables
Users Table
Cashiers Table
Show
10
entries
Search:
ID
Cashier Name
Responsible for
Status
1
Ahmad Saif
Problem 1
Active
2
Mohammed Hamed
Problem 2
Active
3
Liale Ali
Problem 3
Active
Showing 1 to 3 of 3 entries
Previous
1
Next

Admin can Add New Cashier:



Admin can Delete cashier:



Chapter V - SUMMARY, CONCLUSION AND RECOMMENDATION

V.1. Introduction

We will provide background information for the Summary, Conclusion, and Recommendations sections that follow. It outlines the goal of this chapter and gives a summary of the Queue Management System project.

An online program called the Queue Management System was created to improve customer experience and expedite the queuing process in a variety of contexts, including government offices, retail businesses, healthcare institutions, and service centers. Long wait times are to be eliminated, customer annoyance is to be decreased, and resource allocation is to be optimized.
This chapter includes a reflection on the project's salient points, an overview of its accomplishments, conclusions on its influence and success, and suggestions for future development. This chapter provides a thorough assessment of the Queue Management System project and is an invaluable resource for development teams, project managers, and stakeholders.
The following are this chapter's main goals:

1. Provide a summary of the Queue administration System project, emphasizing its main features, including ticket administration, client registration, queue generation, and notifications. We shall list the primary project milestones attained and provide an overview of the entire advancement made.

2. Make inferences about the project's results: We will evaluate how well the Queue Management System satisfies its stated goals and specifications. We'll assess how it affects various parties, such as companies, employees, and clients. We will determine the project's success by considering the system's functionality and the benefits that have been noticed.

3. provide suggestions for upcoming improvements: We will provide recommendations for future improvements and enhancements to the Queue Management System based on our study and review. These suggestions might center on things like more integration, sophisticated reporting and analytics tools, scalability, and flexibility. By offering these suggestions, we hope to direct upcoming choices and encourage the system's ongoing development.

This chapter seeks to give readers a thorough understanding of the Queue Management System project, its results, and its potential for expansion and development in the future by addressing these goals.

V.2. Summary

A succinct synopsis of the Queue Management System project's main attributes, capabilities, and accomplishments is given in the Summary section. It provides an overview of the system's key features and the project's results.

A web application called the Queue Management System was created to completely transform the queuing experience in a variety of contexts. Customers can register, join virtual queues, and get status updates on their queues using an easy-to-use interface. Additionally, the system gives administrators the capabilities they need to effectively manage queues, distribute resources, and keep an eye on service delivery. Several project milestones were accomplished with success. The development of the main features, like client registration, queue generation, and ticket management, was one of these milestones. To guarantee the system's dependability and functionality, extensive testing and quality control procedures were implemented.

Stakeholders have benefited greatly from the Queue Management System's installation. Companies have seen increases in operational efficiency because of more efficient use of resources, shorter wait times, and more efficient service delivery. Because of their improved ability to handle lines, employees are more productive and have better relationships with customers. Consumers now enjoy shorter wait times and greater convenience thanks to a flawless queuing process. The development team, stakeholders, and end users collaborated well, which is why the Queue Management System project was successful. To obtain feedback and incorporate user needs, the project team actively engaged with stakeholders while adhering to agile development processes and industry best practices.

In general, the goals of improving customer experience and revolutionizing the queuing process have been met by the Queue Management System project. Businesses, employees, and consumers have all reacted favorably to the system, providing positive feedback and observable changes.

The accomplishment of the Queue Management System project establishes the groundwork for future improvements and extensions. The system can adapt to the constantly changing demands of organizations and customers by employing advanced analytics, integrating extra communication channels, and incorporating user feedback.

To sum up, the Queue Management System project has been successful in producing a reliable and approachable online application that has completely changed the way queues are handled. The success of the project has been attributed to the team's commitment, observance of best practices, and cooperation with stakeholders. The Queue Management System is a useful tool for streamlining the queuing process and raising customer satisfaction because of its positive effects on stakeholders and its potential for further improvements.

V.3. Conclusion

A thorough evaluation of the Queue Management System project is given in the Conclusion section, which also summarizes the project's results and lessons gained. It assesses the system's overall performance and its effects on different stakeholders by considering the project's goals, accomplishments, and stakeholder input.

It is clear from assessing the Queue Management System project that the system has accomplished its main goals of improving customer satisfaction and

expediting the queuing process. The strategy has effectively tackled the drawbacks of conventional queuing techniques, including protracted wait times, irate patrons, and ineffective resource distribution. The following important criteria are used to evaluate the project's success:

1. Achievement of Project Objectives: The development of basic functionalities, smooth integration with current systems, and provision of an easy-to-use user interface are just a few of the goals that the Queue Management System has achieved. Queue management, resource allocation, and overall customer experience are all improved by the technology.

2. Stakeholder Satisfaction: Positive feedback regarding the Queue Management System has been received from a wide range of stakeholders, including companies, employees, and consumers. Companies have experienced increased service delivery, decreased customer complaints, and increased operational efficiency. Employee productivity has grown, and their relationships with consumers have improved. Shorter wait times, easy notifications, and a smoother queuing experience have all benefitted customers.

3. Positive Impact: The overall efficacy and efficiency of the queuing process have improved thanks to the Queue Management System. Customer satisfaction has increased, waiting times have decreased, and resource allocation has been optimized. The system's streamlined operations have improved service delivery and productivity.

4. Lessons Learned: Several insightful lessons have been discovered during the course of the project. Difficulties faced in the phases of development and execution have yielded valuable insights regarding possible avenues for further enhancement. The project team has acquired significant expertise in managing projects, involving stakeholders, and implementing agile development approaches.

To sum up, the project including the Queue Management System has been a huge success. It has fulfilled expectations from stakeholders, accomplished its goals, and improved queuing procedures in a variety of contexts. The implementation of the system has revolutionized the management of lineups, leading to higher stakeholder satisfaction, improved customer experience, and operational efficiency.

Future improvements and expansions of the Queue Management System project are made possible by its success. Ensuring the system's continuous performance requires constant feedback gathering, integrating developing technology, and adjusting for shifting client needs.
Overall, the Queue Management System is a monument to efficient project management, teamwork, and the use of cutting-edge technology. It has effectively tackled the drawbacks of conventional queuing techniques and offered a strong

remedy that could completely transform the way people queue in a variety of industries.

V.4. Recommendations

The section on recommendations offers ideas for the Queue Management System's future development and enhancement, drawing from the assessment and analysis carried out during the project. These suggestions are meant to serve as a roadmap for next choices and to encourage the system's ongoing development in response to changing demands and expectations.

1. Integration with other Channels: To give consumers more freedom and convenience, it is suggested that the Queue Management System be integrated with other channels of communication. Through the integration of mobile apps or SMS notifications, users may utilize their cellphones to join queues, receive updates, and manage their tickets. More consumers' preferences can be met and the overall user experience improved with this integration.
2. sophisticated Reporting and Analytics: Using the Queue Management System's sophisticated reporting and analytics features can give organizations insightful information that helps them make data-driven decisions. Businesses can further enhance their operations by gathering and evaluating data on consumer behavior, queuing patterns, service efficiency metrics, and resource utilization. These insights can be used to spot bottlenecks, better allocate resources, and anticipate client demands.

3. Scalability and Flexibility: It is advised to concentrate on scalability and flexibility to guarantee the Queue Management System's capacity to manage growing user loads and adjust to shifting business requirements. This may entail making architectural improvements to suit increasing user needs and facilitate future system expansions, such as implementing a modular architecture or utilizing cloud-based infrastructure. The system's scalability and adaptability will facilitate easy integration with new technologies and platforms and allow it to serve a broader user base.

4. Iterative development and continuous user feedback: Promoting and implementing user feedback on a regular basis is essential to enhancing the queue management system. To discover areas for improvement and make sure the system stays in line with user needs, channels for feedback, recommendations, and feature requests from users, employees, and administrators should be established. It is possible for the system to continuously evolve and remain relevant by implementing an iterative development method, in which new features and upgrades are regularly put out depending on user feedback.

5. Integration with Current Systems: It is advised that businesses that have current systems investigate the possibilities for a smooth integration with the Queue Management System. Connecting with appointment scheduling software, customer relationship management (CRM) systems, or other pertinent systems can improve data exchange, optimize workflows, and offer a consistent experience for clients and employees.

6. Improvements in Security and Privacy: As with any digital system, it is crucial to make sure that strong security measures are in place and that consumer privacy is protected. To fix any security flaws, it is advised to patch and upgrade the system on a regular basis. Establishing access restrictions, data anonymization methods, and encryption procedures can help safeguard client information and foster confidence.

By the implementation of these ideas, the functionality, user experience, and overall efficacy of the Queue Management System can be significantly improved. These improvements will help boost customer satisfaction, operational efficiency, and the system's flexibility to accommodate changing demands and trends in queuing management.

REFERENCES

NOMENCLATURE

APPENDICES